Lecture 1 - Introduction

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- The study of signals and system concerns itself with the synthesis and interpretation of signals.
- Digital signals are discretized.
  - Sampling frequencies are used to make discrete signals continuous.
- Systems can be described as functions with a memory (capable of remembering past inputs and outputs).
  - The output of a system depends on its previous state.
  - Bears conceptual similarity to state machines.
  - The input signal to discrete time systems need not be discrete.
- The input to and output from a system is typically denoted as x and y respectively.
  - Discrete systems use n as their independent variable.
    - $n \in \mathbb{N}$.
    - All inputs and outputs for $n < 0$ are defined to be zero.
    - NOTE: n does NOT represent time, it represents the current sample or cycle the system is on.
- We can represent the inputs and outputs of a discrete system at a given cycle n with a table.

At a high-level, we can describe systems with the following diagram:

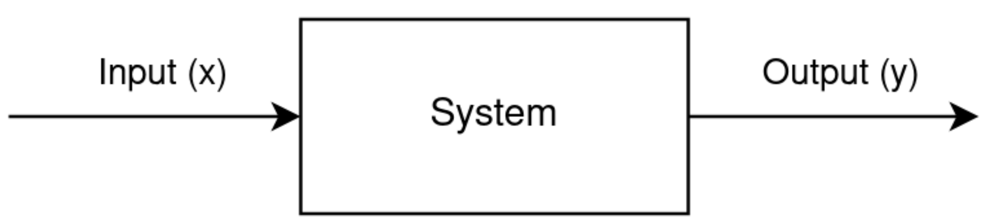

Figure 1: Graphical representation of a system

1 Ex. Compute the outputs to the system $y(n) = \frac{1}{2}x(n) + \frac{1}{2}x(n-1)$ for the following inputs

Note: This system $y(n) = \frac{1}{2}x(n) + \frac{1}{2}x(n-1)$ has a special name, the "Truepoint Moving Average"

1.a $x(n) = \begin{cases} 1, & \text{if } n = 1, 2, 3 \\ 0, & \text{otherwise} \end{cases}$

n	0	1	2	3	4	5
$x(n)$	0	1	1	1	0	0
$y(n)$	0	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	0

1.b $x(n) = \begin{cases} 1, & \text{if } n = 0 \\ 0, & \text{otherwise} \end{cases}$

n	0	1	2	3	4	5
$x(n)$	1	0	0	0	0	0
$y(n)$	$\frac{1}{2}$	$\frac{1}{2}$	0	0	0	0

1.c $x(n) = \cos(n\pi)$

n	0	1	2	3	4	5
$x(n)$	1	-1	1	-1	1	-1
$y(n)$	$\frac{1}{2}$	0	0	0	0	0

2 Ex. Compute the outputs to the system $y(n) = x(n) - y(n-2)$ for the following inputs:

2.a $x(n) = \begin{cases} 1, & \text{if } n = 0 \\ 0, & \text{otherwise} \end{cases}$

n	0	1	2	3	4	5
$x(n)$	1	0	0	0	0	0
$y(n)$	1	0	-1	0	1	0

2.b $x(n) = \cos(\frac{1}{2}\pi n)$

n	0	1	2	3	4	5
$x(n)$	1	0	-1	0	0	1
$y(n)$	1	0	-2	0	3	0

3 Ex. Compute the outputs to the system $y(n) = x(n) + \frac{1}{2}y(n-1)$ where $x(n) = \begin{cases} 1, & \textbf{if } n = 0 \\ 0, & \textbf{otherwise} \end{cases}$

n	0	1	2	3	4	5
$x(n)$	1	0	0	0	0	0
$y(n)$	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$

4 Ex. Compute the outputs to the system $y(n) = x(n) + x(n-1) - y(n-1)$ where $x(n) = n + 1$

n	0	1	2	3	4	5
$x(n)$	1	2	3	4	5	6
$y(n)$	1	2	3	4	5	6

- A system (such as the one given in exercise 2.b) which diverges to $\pm\infty$ as the number of cycles increases is called unstable.
  - Relates to a concept known as resonant frequencies.
- A system (such as the one given in exercise 4) for which $y(n) = x(n)$ is called an identity system.
- All discrete systems can be represented by a Difference Equation (DEQ)
  - DEQs represent systems as a linear combination of the system's current input, past inputs, and past outputs.
- The DEQ for any discrete system can be generalized by the formula:
  - $y(n) = \sum_{k=0}^{N} \alpha_k x(n-k) + \sum_{k=1}^{M} \beta_k y(n-k), \quad N, M \in \mathbb{N}$
    * Note that the lower bound for the sums of the $y(n-k)'s$ is 1.
    * Also note that each sum is finite.

5 Ex. Represent the system $y(n) = \frac{1}{2}x(n) - \frac{1}{2}x(n-1)$ as a Difference Equation (DEQ)

Selecting $N = 1, M = 0, \alpha_k = \begin{cases} \frac{1}{2} \text{ if } k = 0 \\ -\frac{1}{2} \text{ if } k = 1 \\ 0, \text{ otherwise} \end{cases}$, and $\beta_k = 0,\ k \in \mathbb{N}$, we can see that:

$$\begin{aligned} y(n) &= \sum_{k=0}^{N} \alpha_k x(n-k) + \sum_{k=1}^{M} \beta_k y(n-k) \\ &= \frac{1}{2}x(n) - \frac{1}{2}x(n-1) \end{aligned}$$

Thus, we have successfully represented our system using the general DEQ form.